

LINUX CHEAT SHEET

Kali Linux / Debian • Virtual Cybersecurity Lab Reference

Purpose: Quick-reference commands for Linux administration, networking, troubleshooting and authorized cybersecurity lab exercises.

1. System Information

COMMAND	PURPOSE
<code>uname -a</code>	Display kernel, architecture, hostname and system information.
<code>hostname</code>	Display the system hostname.
<code>hostnamectl</code>	View or manage hostname and system identification details.
<code>cat /etc/os-release</code>	Display Linux distribution and release information.
<code>uptime</code>	Show system uptime and load information.

2. User Information

COMMAND	PURPOSE
<code>whoami</code>	Show the current username.
<code>id</code>	Display user ID, group ID and memberships.
<code>who</code>	Show users currently logged in.
<code>w</code>	Show logged-in users and their activity.

3. Network

COMMAND	PURPOSE
<code>ip addr</code>	Display interfaces and IP addresses.
<code>ip route</code>	Display the routing table.
<code>ip link</code>	Display/manage network interfaces.
<code>ping</code>	Test IP connectivity using ICMP.
<code>ss -tulnp</code>	List listening TCP/UDP sockets with process information.

4. DNS

COMMAND	PURPOSE
<code>cat /etc/resolv.conf</code>	View configured DNS resolver settings.
<code>nslookup example.com</code>	Query DNS records using nslookup.
<code>dig example.com</code>	Perform detailed DNS queries and inspect responses.

5. Files & Directories

COMMAND	PURPOSE
<code>pwd</code>	Print the current working directory.
<code>ls</code>	List files and directories.
<code>ls -la</code>	List all files, including hidden files, with details.
<code>cd</code>	Change directory.

COMMAND	PURPOSE
<code>mkdir</code>	Create a directory.
<code>touch</code>	Create a file or update its timestamp.
<code>cp</code>	Copy files/directories.
<code>mv</code>	Move or rename files/directories.
<code>rm</code>	Remove a file; use recursive options carefully.

6. File Viewing

COMMAND	PURPOSE
<code>cat file.txt</code>	Print file contents.
<code>less file.txt</code>	View a file interactively, page by page.
<code>head file.txt</code>	Show the beginning of a file.
<code>tail file.txt</code>	Show the end of a file.
<code>tail -f file.log</code>	Follow a log and display new lines as they appear.

7. Searching

COMMAND	PURPOSE
<code>find / -name filename</code>	Search the filesystem for a matching filename; permissions may cause errors.
<code>grep "text" file.txt</code>	Search for matching text within a file.

8. Permissions

COMMAND	PURPOSE
<code>ls -l</code>	Display permissions, ownership and metadata.
<code>chmod</code>	Change file/directory permissions.
<code>chown :</code>	Change file/directory ownership.

9. Processes

COMMAND	PURPOSE
<code>ps aux</code>	Display running processes.
<code>top</code>	Interactive real-time process/resource monitor.
<code>htop</code>	Interactive process viewer; may need installation.
<code>kill</code>	Send a termination signal to a process.

10. Services

COMMAND	PURPOSE
<code>systemctl status</code>	View systemd service status.
<code>systemctl start</code>	Start a service.
<code>systemctl stop</code>	Stop a service.
<code>systemctl restart</code>	Restart a service.

11. Package Management — Kali/Debian

COMMAND	PURPOSE
<code>sudo apt update</code>	Refresh package repository metadata.
<code>sudo apt upgrade</code>	Upgrade installed packages.
<code>sudo apt install</code>	Install a package.
<code>sudo apt remove</code>	Remove an installed package.

12. Useful Cybersecurity Commands

COMMAND	PURPOSE
<code>ip addr</code>	Identify local interfaces and IP addresses.
<code>ss -tulnp</code>	Inspect listening services and associated processes.
<code>ping</code>	Check basic host reachability.
<code>traceroute</code>	Trace the network path to a host.
<code>nmap</code>	Perform basic host/port discovery in an authorized lab.
<code>nmap -sV</code>	Attempt service/version detection on discovered ports.
<code>tcpdump</code>	Capture and inspect packets from the command line.
<code>wireshark</code>	Capture and analyze packets using a graphical interface.

LAB SAFETY & GOOD PRACTICE

- Use scanning and packet-capture commands only against systems you own or are explicitly authorized to test.
- Keep intentionally vulnerable machines such as Metasploitable 2 on an isolated host-only/internal network.
- Review commands before destructive operations such as `rm`, `chmod`, `chown`, `kill` or service management.
- Replace placeholders such as `<IP>`, `<PID>`, `<service>` and `<package>` with appropriate values.